NREPESH JOSHI

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PROFESSIONAL EXPERIENCE

TACKLE AI, Schaumburg, IL (Oct 2020 - Present)

Second hire on the ML team, reporting to the CTO. Own the stack end to end: data, models, deployment, and monitoring, built to run on-premise for the country's largest law firms so the medical records it processes never reach a hosted model.

Machine Learning Team Lead (Mar 2023 - Present)

- Hired and now lead a team of 5 ML engineers, driving 90% of company revenue in a critical quarter and sustaining 60% since.
- Fine-tuned and benchmarked TackleVision's OCR engine to 86.1 on olmOCR-bench, the highest independently reproducible score published, across 8,413 tests on 1,403 real-world pages.
- Shipped the company's first production LLM systems, replacing LangChain and vector DBs with custom similarity search for a 25x speed boost.
- Built the team's ML platform and shared packages for OCR, file handling, and agentic LLM templates, ending duplicated work across projects.
- Built an autolabeler that uses idle overnight GPU capacity to pre-label datasets, and applied synthetic data augmentation to lift object detection performance 30%.
- Stood up air-gapped on-premise infrastructure at a law client's site, building and deploying the full ML stack inside an isolated environment.
- Scaled the computer vision and NLP models behind automated redaction to 250K+ pages per day in production, supporting class action litigation including opioid crisis productions.
- Own the production bar for AI-assisted code, auditing latency, live-data safety, and output correctness; cut per-document processing from 2 hours to 15 minutes on less memory.
- Processed 3M+ PHI-bearing patient records through HIPAA and SOC 2 Type II compliant on-premise pipelines, classifying document types with LayoutLMv3 and vector retrieval.
- Built an asynchronous inference backend on gRPC and WebSockets, orchestrating multimodal LLMs on vLLM through Redis and Celery to return structured OCR JSON in real time.

Machine Learning Engineer (Oct 2020 - Mar 2023)

- Trained the bounding-box, instance, and semantic segmentation models that carried the platform through its seed round.
- Built an optimized image retrieval model using feature-extractor embeddings and custom LSH vector spaces, holding 92% accuracy across a 1M+ image corpus.
- Ported a Kinesis video pipeline from Python to Java, taking a lagging feed to a real-time stream.

- Consolidated a multi-model inference pipeline into a single ensemble call, speeding up inference by 75%.
- Fine-tuned spaCy DistilBERT for Named Entity Recognition, reaching an F1 of 97% on production extraction.
- Migrated spaCy NLP models to Triton Server, cutting vRAM usage 10x and freeing capacity for more model deployments.
- Reduced manual labeling workload 85% through unsupervised clustering (HDBSCAN), dimensionality reduction, and data standardization.
- Established CI/CD pipelines for deploying ML models via gRPC and REST APIs on AWS using ECS, ECR, and Lambda.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA **Jan 2026 - Present**

Master of Science in Computer Science; Machine Learning Specialization

FOURTH BRAIN, Machine Learning Academy, Online **Nov 2021 - Feb 2022**

MLOps and Systems Program, backed by Andrew Ng

KNOX COLLEGE, Galesburg, IL **Sept 2016 - June 2020**

Bachelor of Science; Double Major in Computer Science & Physics

Magna Cum Laude ~ Cumulative GPA 3.77/4.0 ~ CS Major GPA 4.0/4.0

Scholarships: Ellen Browning Scripps, Service & Leadership and Knox Grant

COMMUNITY INVOLVEMENT

- **Speaker, AI Conference for a Prosperous Nepal.** Presented "AI & ML in Government Projects," covering practical AI applications for societal benefit and addressing data and computational challenges.
- **Chicago In-Context MCP Mini-Hackathon 2025, 2nd Place Winner.** Developed FastMCP server enabling LLM tool calling to query Chicago building heights and recommend offline map downloads based on urban density analysis.

SKILLS

- **Programming Languages:** Python, Java, C#
- **Machine Learning Frameworks:** TensorFlow, Keras, PyTorch, Hugging Face, spaCy
- **LLMs and Generative AI:** vLLM, LangChain, LlamaIndex, OpenAI, RAG, vector search, agentic workflows
- **MLOps and Monitoring:** MLflow, Prometheus, Grafana, Label Studio, Triton Server
- **Web Frameworks and Databases:** Flask, FastAPI, Nginx, S3, MySQL, MongoDB, Redis, Milvus
- **Infrastructure:** On-premise and air-gapped deployment, Docker, Kubernetes, Linux, AWS, GCP